Kha Pham

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Skills

Languages: Python, SQL, SAS, Java, Kotlin, JavaScript/TypeScript, Bash.
Backend: Flask, REST APIs, Webhooks, Gunicorn, MongoDB, Redis.
DevOps & Cloud: GitLab CI, GitHub Actions, Docker, Docker Compose, Ansible, SonarQube, Trivy, Sonatype Nexus, AWS (Lambda, API Gateway, EC2), GCP (App Engine), Linux, Tailscale.
Observability: Grafana, Loki, Graylog, OpenTelemetry, Prometheus, Dynatrace.

Relevant Experience

Software Developer - Bureau of Labor Statistics — Aug. 2024 - Present

Washington, D.C., USA

- Built and maintained GitLab CI/CD pipelines automating SSH/rsync deployments of 3 services to acceptance servers on an ad-hoc basis.
- Integrated unit testing, SonarQube static analysis, and Trivy vulnerability scanning as quality gates across acceptance and production pipelines in an air-gapped environment backed by internal Sonatype Nexus for a service modernization project.
- Created a VS Code Remote-SSH setup using per-user sshd instances on custom ports, enabling modern IDE workflows on servers that prohibit port forwarding.
- Contributed to agency-wide SAS-to-Python modernization by writing and testing Python replacements for legacy workflows and building a SAS codebase inventory tool to support migration planning.
- Mitigated federal shutdown data gaps by engineering a simulation environment to mock missing Consumer Expenditure survey data, enabling continued program office operations.
- Prototyped a centralized Graylog logging dashboard across 4 microservices, replacing file-based logging to improve observability ahead of the team's AWS migration.

Site Reliability Engineer, Developer Internship - Royal Bank of Canada — May 2023 - Aug. 2023

Toronto, Ont., Canada

- Developed a health monitoring tool across 6-7 Dynatrace clusters, dispatching email and Slack alerts when configurable, application-specific performance thresholds were exceeded.
- Automated team permission management on RBC's internal GitHub platform using Python and Ansible, allowing access rights, groups, and tags to be updated across all team members via a single config file.

Projects

Homelab GitOps Platform — Mar. 2026 - Present

github.com/khathepham/homelab

- Built a GitOps pipeline in GitHub Actions that detects changed Docker Compose stacks from each merged PR and deploys them across 2 hosts running 17 self-hosted services.
- Secured CI access with ephemeral Tailscale credentials and Infisical-managed secrets; monitor services with Grafana, Loki, and Uptime Kuma.

.devHacks 2025 Event Services — Jan. 2025 - Feb. 2025

- Built three containerized Flask microservices (registration, check-in, teams) using webhooks, a Notion-backed datastore, and QR-code tickets for a 250-attendee hackathon.
- Automated releases with GitHub Actions, publishing images to GHCR and redeploying via Portainer webhooks.

SAS Audit — Jan. 2026 - Present

github.com/khathepham/sasaudit

- Built a Python static-analysis CLI that inventories SAS codebases (cross-file dependencies, macros, PROC usage, Oracle calls) and exports multi-sheet Excel reports to support migration planning; covered by 37 pytest tests.

Leadership and Community Experience

CUSEC: Canadian University Software Engineering Conference — Apr. 2022 - Present

Chair — *Feb. 2024 - Present*

- Selected and onboarded a 25+ person organizing team from a 90+ candidate pool, conducting structured interviews to staff all conference committees.
- Led a 25+ person team to deliver a 3-day, 350+ attendee national conference on time and on budget, stepping in across under-resourced teams to maintain momentum.

Director of External Affairs - .devClub — June 2023 - May 2024

University of Manitoba, Winnipeg, Man., Canada

- Secured industry sponsorships funding .devHacks, a 300+ student hackathon, through targeted outreach and negotiation with technology companies.
- Delivered a workshop on API development and Flask to 100+ upper-year students, directly increasing participation in the club's subsequent Battlesnake tournament.

Education

B.S. Computer Science - University of Manitoba — Sept. 2021 - Apr. 2024

Winnipeg, Man., Canada

- Dean's Honour List, Fall 2022
- Relevant Coursework: Algorithms, Artificial Intelligence, Distributed Computing, Databases, Software Engineering, Data Mining, Data Science
- External Executive, .devClub; Executive, UM Pep Band; Member, Rotaract Club